

EDA REPORT ANALYSIS

Executive Summary

Analyzed 114,000 charging sessions, 8,000 customers, 65 operational stations, 18 districts, and 98 proposed EV charging projects across Bengaluru. The study evaluates demand, customer behavior, network performance, investment feasibility, and expansion opportunities.

Key KPIs

- Total Charging Sessions: 114,000
- Registered Customers: 8,000
- Operational Stations: 65
- Districts Covered: 18
- Proposed Projects: 98
- Total Revenue: 4.97 Crore
- Total Energy Delivered: 28.60 Lakh kWh
- Average Revenue per Session: 435.91
- Average Sessions per Customer: 14.25

Dataset Overview

Five datasets were used: charging_sessions, customers, charging_stations, districts, and roi_bengaluru_districts. The datasets are connected through customer_id, station_id, and district attributes.

Charging Session Analysis

The network processed 114,000 charging transactions in approximately five months. Average energy delivered per session was about 25.09 kWh. Average customer spend per charging session was 435.91. This indicates strong utilization of the charging network.

Customer Analysis

Customer base: 8,000 registered users.
Low-Mid segment: 3,990 customers (~49.9%).
Mid-Range segment: ~35% of users.
High-income segment: ~15% of users.
High-income customers contribute disproportionately high revenue.

District Analysis

High-demand districts include Indiranagar, Whitefield, Koramangala, Jayanagar, and Rajajinagar. These districts exhibit the strongest EV adoption potential and should remain expansion priorities.

Operator Analysis

The charging market is competitive among Tata Power, ChargeZone, BPCL Pulse, Ather Grid, and Statiq. Revenue contribution is relatively balanced, indicating opportunities for market-share growth.

ROI Analysis

98 proposed projects were evaluated. Priority Distribution: • High Priority: 54 projects • Medium Priority: 7 projects • Low Priority: 37 projects Average Payback Period: 4.61 years Average 5-Year

ROI: 41.43%

Financial Analysis

Total Planned CAPEX: 139.22 Crore Annual Revenue Potential: 46.29 Crore Annual EBITDA Potential: 32.39 Crore The investment portfolio demonstrates attractive economics for long-term EV infrastructure expansion.

Subsidy Analysis

55 projects qualify for government subsidies. Total subsidy support exceeds ₹7.81 Crore. Subsidies materially improve project payback and ROI.

Top Investment Opportunities

Top-performing districts based on ROI: • Marathahalli • HSR Layout • Banashankari • Yelahanka • Whitefield These locations offer the strongest investment potential.

Management Recommendations

1. Prioritize all high-priority projects. 2. Expand fast-charging infrastructure in premium districts. 3. Target Mid-Range and High-income customer segments. 4. Reassess low-priority projects before capital deployment. 5. Utilize available government subsidies aggressively.

Executive Insights

Insight 1: The existing network generated approximately ₹4.97 Crore in revenue. Insight 2: More than half of proposed projects are classified as high-priority investments. Insight 3: Premium Bengaluru districts remain the strongest EV adoption centers. Insight 4: Subsidy support significantly improves investment feasibility and reduces risk.

DASHBOARD OVERVIEW

Dashboard is created in ms-excel.

